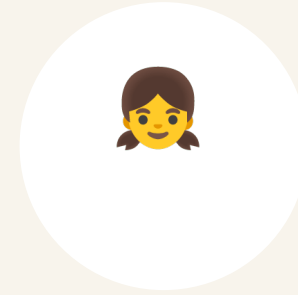


From predictive model to responsible developmental navigation

AIM AI/ML Capstone Project

Using population data to explore whether machine learning can support earlier developmental review while avoiding diagnostic overreach.



developmental review

PROBLEM

The business problem: demand exceeds specialist capacity

CURRENT REALITY

Families ask for developmental assessment when the system is already crowded.

Specialist slots are scarce and waitlists grow quickly
Families may not know whether concern warrants urgent action
Missions and clinics need a lightweight intake logic
Current recognition patterns may miss some children, especially girls

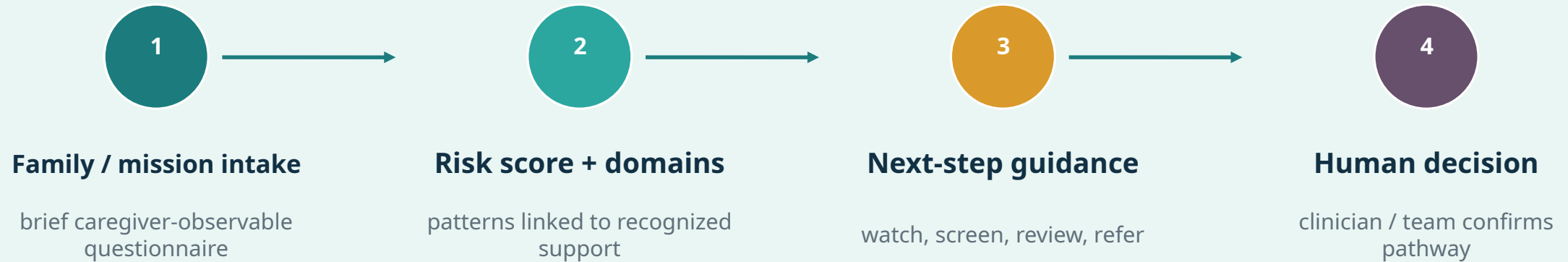
Many inquiries

Limited triage

Specialist slots

Opportunity: use ML as an intake support layer, not as a diagnostic gatekeeper.

Proposed use case: developmental navigation, not diagnosis



Positioning guardrail

The model supports prioritization and conversation. It does not diagnose, deny services, or replace professional judgment.

Proof of concept: useful signal in public population data

7,485

children ages 3–5

27.1%

positive composite outcome

102

initial candidate predictors

4.07%

maximum missingness

What was modeled?

A composite outcome of recognized developmental concern or support, based on diagnoses, developmental / behavioral treatment, early intervention, special education, and related support indicators.

Why this matters commercially

Can be explored before collecting local clinical data
Low-cost way to test feasibility and define validation needs
Creates a research bridge toward clinic, missions, and CHLD workflows

Core caution: the model predicts recognition/support in a dataset, not true developmental need.

Business value: focus scarce attention where it may matter most

1 Prioritize

Sort inquiries into practical next-step levels

2 Standardize

Use consistent intake logic across clinics and missions

3 Explain

Surface developmental domains driving concern

4 Learn

Build evidence for local validation and service planning

ROI is not only financial

The near-term return is operational: fewer ad hoc replies, more structured intake, clearer routing, and better evidence for resource planning. Financial ROI comes later through validated workflow efficiency and scalable service design.

Candidate KPIs: intake completion rate • time-to-next-step • referral uptake • fairness gaps • clinician review burden

Key result: a reduced model preserved performance

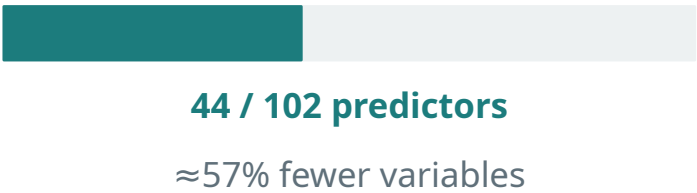
Final cross-validated comparison

Model	Features	ROC-AUC	PR-AUC	F1	Sens.	Acc.
A: Full	102	.782	.662	.510	.381	.802
B: Domain	38	.766	.643	.500	.373	.798
C: Data-driven	44	.785	.664	.518	.390	.803

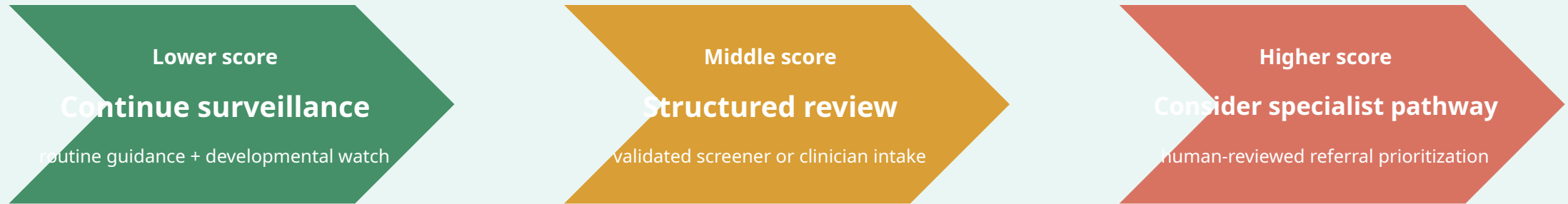
Executive interpretation

The best reduced model used 44 predictors, fewer than half of the original feature set, and slightly outperformed the full model. This suggests a practical path toward a shorter intake tool, but not yet a deployable clinical product.

Feature burden



How it would operate: thresholds create action zones



Why not one fixed cutoff?

A 0.50 threshold maximized caution but missed many positive cases
Lower thresholds detect more children but increase review workload
The “right” threshold depends on downstream action and capacity

Business principle

Do not equate “flagged” with “diagnosed.”
The model should trigger a proportionate next step, not a final decision.

Risks are manageable only with governance

Risk

False reassurance

A child who needs review may be missed

False positives

Families may be routed into unnecessary review

Recognition bias

Labels may reflect who gets noticed or supported

Privacy

Pediatric intake data requires strict minimization

Over-reliance

Users may treat the output as a diagnosis

Mitigation

→ Use as triage support, never rule-out diagnosis

→ Proportionate next steps, not automatic specialist referral

→ Monitor sex, age, language, and socioeconomic performance

→ Local-first research, de-identification, minimal fields

→ Human oversight and clear user-facing disclaimers

Governance rule: the greater the consequence of the output, the stronger the required human review.

Roadmap: from capstone to responsible pilot

0–3 months

Research protocol

ethics board submission
local data dictionary
workflow and consent design

3–6 months

Local validation dataset

clinic / mission intake
human-reviewed outcomes
calibration + fairness checks

6–12 months

Pilot tool

local-only prototype
dashboard for review
threshold and KPI monitoring

Build criterion

Proceed only if local validation shows acceptable calibration, subgroup sensitivity, clinician usability, and manageable review burden.

Recommended executive decision

Do not deploy as a clinical screener yet.

Use the capstone as the evidence base for a local validation study and a privacy-first intake prototype.

Next ask

- Approve research-protocol preparation
- Define local validation outcomes
- Design a brief caregiver intake workflow
- Pre-specify fairness and safety metrics
- Review after first local dataset is collected

Strategic conclusion

The opportunity is not to replace developmental assessment. It is to make the path toward assessment more structured, equitable, and evidence-guided.